

Fixed- h_0 De-Aliasing after Frozen Averaging: Exact Evaluation Norms, Christoffel Costs, and a Sharp Localization Obstruction

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Abstract

A frozen-parameter or frequency-averaged estimate is not yet an estimate at one prescribed physical shift h_0 . The missing operation is evaluation on the actual shift-profile subspace. We give its exact finite-dimensional cost.

Let $A : E \rightarrow \mathbb{C}^I$ be the declared observation map, let $\mathbf{\Omega} = \text{diag}(p_i)$ with $p_i \geq 0$, and let $\ell(x) = b_0^* x$ be the target functional. With $G = A^* \mathbf{\Omega} A$, a finite localization inequality

$$|\ell(x)|^2 \leq K \|Ax\|_{\mathbf{\Omega}}^2 \quad (x \in E)$$

exists if and only if $b_0 \in \text{ran } G$, equivalently ℓ annihilates $\ker A_{\mathbf{\Omega}}$, where $A_{\mathbf{\Omega}} = \mathbf{\Omega}^{1/2} A$. In that case the sharp constant is

$$K = b_0^* G^\dagger b_0.$$

For a profile space $V = \text{ran } B$, this becomes the Christoffel evaluation cost

$$K_V(h_0) = b(h_0)^* (B^* \mathbf{\Omega} B)^\dagger b(h_0).$$

The formula separates three regimes. The unrestricted profile space costs exactly $1/p_{h_0}$, reproducing the positivity-only localization loss. Constants cost 1. A d -character subspace under uniform averaging costs exactly d , independent of the ambient number of shifts. Conversely, if the target does not annihilate the observation kernel, no finite averaged estimate can recover it.

This is a sharp conditional bridge, not a fixed-shift arithmetic estimate. It advances the interface only after the actual TPC shift profile is proved to lie in a controlled subspace. We prove no such regularity, no L2 fixed-power saving, no parity breakthrough, and no prime-pair or twin-prime theorem.

Keywords: prescribed shift; localization; Christoffel cost; pseudoinverse; averaged estimate; de-aliasing.

Physical convention. The target $h_0 \neq 0$, averaging weights, active shift support and profile space are declared before optimization. A profile space fitted after seeing the target is not a localization theorem.

1 The missing return from averages

TPC-46 exhibits the sharp positivity cost $\sum_h \omega_h / \omega_{h_0}$ for recovering one shift from a weighted average with no regularity [4]. Later frozen and spectral routes improve averaged or model estimates, but TPC-MVP1 requires an actual prescribed physical shift [1]. TPC-113 fixes physical conditioning, and TPC-114 identifies the residual/native Gram and its cancellation kernel [2, 3]. The present paper isolates the exact evaluation norm after those maps.

Let E be a finite-dimensional Hilbert space and

$$A : E \rightarrow \mathbb{C}^I, \quad \Omega = \text{diag}(p_i : i \in I), \quad p_i \geq 0.$$

Write

$$\|z\|_\Omega^2 = z^* \Omega z, \quad A_\Omega = \Omega^{1/2} A, \quad G = A^* \Omega A = A_\Omega^* A_\Omega.$$

The target is $\ell(x) = b_0^* x$.

2 The exact localization theorem

Theorem 2.1 (Sharp evaluation through a semidefinite observation). *There exists a finite K such that*

$$|b_0^* x|^2 \leq K x^* G x \quad (x \in E)$$

if and only if

$$b_0 \perp \ker G, \quad \text{equivalently} \quad b_0 \in \text{ran } G.$$

When this holds, the least possible constant is

$$K_{\text{loc}}(A, \Omega; b_0) = b_0^* G^\dagger b_0.$$

If it fails, the sharp constant is $+\infty$.

Proof. Since $G = A_\Omega^* A_\Omega$, $\ker G = \ker A_\Omega$. If b_0 does not annihilate this kernel, some $x \in \ker G$ has $b_0^* x \neq 0$, so no finite inequality is possible.

Assume $b_0 \in \text{ran } G$. On $(\ker G)^\perp$, put $y = G^{1/2} x$. Then

$$b_0^* x = (G^{\dagger/2} b_0)^* y.$$

Cauchy–Schwarz gives

$$|b_0^* x|^2 \leq \|G^{\dagger/2} b_0\|^2 \|G^{1/2} x\|^2 = (b_0^* G^\dagger b_0)(x^* G x).$$

Equality holds for $y = G^{\dagger/2} b_0$, equivalently along $x = G^\dagger b_0$, unless $b_0 = 0$, when both sides vanish. Thus the constant is sharp. $\square$

Remark 2.2 (Kernel condition, not invertibility). The observation need not be injective on all of E . Only the target must vanish on its invisible directions. Requiring a global inverse can therefore overstate the cost, while ignoring the kernel can assert an impossible de-aliasing.

Proposition 2.3 (Hilbert-valued extension). *Let $\mathcal{H}$ be any Hilbert space and replace A by $A \otimes I_{\mathcal{H}}$, and b_0^* by $b_0^* \otimes I_{\mathcal{H}}$. Under the condition of [theorem 2.1](#),*

$$\|(b_0^* \otimes I_{\mathcal{H}})X\|_{\mathcal{H}}^2 \leq K_{\text{loc}} \|(A_\Omega \otimes I_{\mathcal{H}})X\|^2,$$

with the same sharp constant.

Proof. Apply the scalar inequality to the scalar products of the $\mathcal{H}$ -valued coefficients against arbitrary unit vectors, or diagonalize G and sum the componentwise inequalities. Sharpness follows by tensoring a scalar extremizer with a unit vector. $\square$

3 Profile spaces and Christoffel cost

Let H be the finite shift set, let $p_h \geq 0$ with $\sum_h p_h = 1$, and let $B : \mathbb{C}^d \rightarrow \mathbb{C}^H$ describe a declared profile space:

$$(Bc)(h) = b(h)^* c.$$

Evaluation at h_0 is $b(h_0)^* c$.

Corollary 3.1 (Exact profile-space localization). *Put*

$$G_V = B^* \Omega B, \quad \Omega = \text{diag}(p_h : h \in H).$$

If $b(h_0) \in \text{ran } G_V$, then

$$|(Bc)(h_0)|^2 \leq K_V(h_0) \sum_{h \in H} p_h |(Bc)(h)|^2,$$

where the sharp constant is

$$K_V(h_0) = b(h_0)^* G_V^\dagger b(h_0).$$

If $b(h_0) \notin \text{ran } G_V$, no finite constant exists.

Proof. Apply [theorem 2.1](#) with $A = B$ and $b_0 = b(h_0)$. □

Proposition 3.2 (Three exact regimes). *The following constants are sharp.*

(i) For the unrestricted profile space $\mathbb{C}^H$,

$$K_{\mathbb{C}^H}(h_0) = \frac{1}{p_{h_0}}$$

when $p_{h_0} > 0$; it is infinite when $p_{h_0} = 0$.

(ii) For the constant profile space, $K_{\text{const}}(h_0) = 1$.

(iii) Let $H = \mathbb{Z}/N\mathbb{Z}$, use uniform weights $1/N$, and let V be the span of any d distinct characters. Then

$$K_V(h_0) = d \quad \text{for every } h_0.$$

Proof. For (i), $B = I$, $G_V = \Omega$, and the target is e_{h_0} . For (ii), $B = \mathbf{1}$, so $G_V = \sum_h p_h = 1$. For (iii), character orthogonality gives $B^*(N^{-1}I)B = I_d$, while the evaluation vector $b(h_0)$ has d unit-modulus entries. Hence $b(h_0)^* b(h_0) = d$. □

Remark 3.3 (The positivity cost is genuinely sharp). Writing $p_h = \omega_h / \sum_t \omega_t$, part (i) gives

$$K_{\mathbb{C}^H}(h_0) = \frac{\sum_t \omega_t}{\omega_{h_0}},$$

exactly the TPC-46 positivity cost. It cannot be improved from an averaged inequality alone. Any improvement must prove that the actual profile lies in a smaller declared space, as in parts (ii) or (iii), or establish another explicit regularity inequality.

4 Conditional fixed-shift transfer

Theorem 4.1 (What an averaged arithmetic estimate would imply). *Let $F_X(h) \in \mathcal{H}_X$ be the literal physical shift profile. Assume:*

(a) F_X belongs to a declared Hilbert-valued profile space $V_X \otimes \mathcal{H}_X$;

(b) the averaged estimate

$$\sum_h p_{X,h} \|F_X(h)\|_{\mathcal{H}_X}^2 \leq E_X$$

holds on the same active support and normalization; and

(c) the target evaluation annihilates the observation kernel.

Then

$$\|F_X(h_0)\|_{\mathcal{H}_X}^2 \leq K_{V_X}(h_0) E_X.$$

If $K_{V_X}(h_0) = X^{\lambda+o(1)}$, localization spends λ in an energy ledger and $\lambda/2$ in the corresponding amplitude ledger.

Proof. Apply [proposition 2.3](#) and [corollary 3.1](#). The exponent statement follows by taking square roots when converting squared norms to amplitudes. $\square$

Remark 4.2 (Strict endpoint accounting). The localization cost is charged exactly once, after the native Gram and before the prescribed-shift conclusion. If the route uses the amplitude-normalized physical ledger of TPC-MVP1, the accumulated loss, including $\lambda/2$, must remain strictly below $1/400$. An endpoint equality is not a positive reserve.

5 Sharp invisibility obstruction

Proposition 5.1 (Indistinguishable profiles). *If $x_1 - x_2 \in \ker A_{\Omega}$ but $\ell(x_1) \neq \ell(x_2)$, then no procedure using only the weighted observation $A_{\Omega}x$ can recover the target on both profiles. In particular, no finite localization inequality exists.*

Proof. The observations of x_1 and x_2 coincide, whereas their target values differ. Their difference is the kernel witness in [theorem 2.1](#). $\square$

For example, $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$ observes only $x_1 + x_2$. Evaluation of the first coordinate does not annihilate $\ker A = \text{span}\{(1, -1)\}$, so it is impossible. Evaluation of the observed sum itself has sharp cost 1.

6 Finite exact certificate and verdict

The script `experiments/tpc115_localization_certificate.py` checks with exact rational arithmetic:

1. unrestricted weighted costs 6, 3, 2 for weights 1/6, 2/6, 3/6;
2. constant-profile cost 1;
3. Walsh-character subspace costs $d = 1, 2, 4, 8$ on eight shifts; and
4. the invisible and visible functionals for $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$.

The certificate validates the finite formulas, not an arithmetic profile hypothesis.

Status	Required certificate	Consequence
Go	The actual profile belongs to V_X , the averaged estimate is physical, and $K_{V_X}(h_0)$ leaves strict endpoint reserve	Apply theorem 4.1 and continue to high-frequency reassembly.
Stop observation	Target evaluation is nonzero on the actual observation kernel	No de-aliasing from this average is possible; retain more labels or change the observation.
Open	Only an averaged bound is known, with no verified profile space	Pay the full positivity cost or make no fixed- h_0 claim.

Established. [Theorems 2.1](#) and [4.1](#), [propositions 5.1](#) to [2.3](#), and [corollary 3.1](#) give the exact evaluation norm, its Hilbert-valued extension, and a sharp obstruction. These are L0 finite-dimensional theorems. Attaching them to the literal fixed-shift profile is an explicit L1 conditional interface.

Not established. We do not prove that the actual TPC profile lies in a low-dimensional character or regularity space. We prove no bound for its Christoffel cost, no new averaged arithmetic estimate, and no prescribed-shift saving. There is no L2 advance, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem. Nothing is specialized to $h_0 = 2$.

References

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